

Question

The following chemical processes exist:

1. Passing AsH_3 into the $\text{HF} - \text{SbF}_5$ system can yield a highly symmetrical anion and cation.
2. At 203 K, AlCl_3 can react with N_2F_4 to produce a fluoride with one C_2 axis and one plane of symmetry. This reaction can be understood as N_2F_4 first undergoing disproportionation, and then reacting with AlCl_3 .
3. At high temperature and in the presence of H_2 , chromium(III) oxide can react with graphite to obtain a compound with a chromium content of 86.7%.
4. At 242 K, $[\text{NMe}_4]\text{I}$ can react with XeF_2 in a molar ratio of 1 : 2.
5. The carbonyl compound $\text{O}_2\text{NOMn}(\text{CO})_5$ is a stable nitrate complex that decomposes upon heating to produce a common nitrate and release gas.

Regarding the chemical equations of the above reactions, assuming the coefficients are balanced to the simplest integer ratio, the following statements are correct:

- A. All other options are incorrect
- B. The anion and cation produced in Process 1 possess the same symmetry.
- C. Chemical equation for process 2, the sum of coefficients of all species is less than 10.
- D. The chemical equation for Process 3, the sum of the coefficients of all species is less than 25.
- E. The stoichiometric coefficients sum of all species in the chemical equation for Process 4 is a prime number.

F. The chemical equation for process 5, the sum of the coefficients of all species is a prime number.

Answer

Correct Answer: **F**

Detailed Explanation

Process 1: The anion of the $\text{HF} - \text{SbF}_5$ system is easily conceived as SbF_6^- ; the cation has high symmetry. Considering the Lewis acid-base reaction of non-aqueous solvents, it is likely to be AsH_4^+ . Balancing the equation:



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The anion and cation in the product do not have the same symmetry. Option B is incorrect.

Process 2: First consider the disproportionation reaction of N_2F_4 ; the +2 valence disproportionates to a stable valence state, generally +3 NF_3 and 0 valence N_2 ; then consider a fluoride with a C_2 axis and a plane of symmetry, the only thing that comes to mind is N_2F_2 . Therefore, in the reaction of NF_3 with AlCl_3 , nitrogen is reduced, so chlorine can only be oxidized to chlorine gas, while producing the fluorinated product AlF_3 . Therefore, the equation is balanced as follows:

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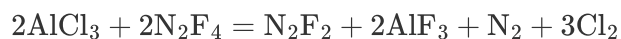
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+2 valence N disproportionates to a stable valence state, generally +3 NF_3 and 0 valence N_2

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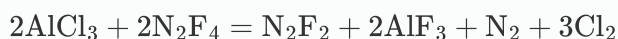
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A fluoride with a C_2 axis and a plane of symmetry, the only thing that comes to mind is N_2F_2



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The sum of the coefficients is 11, which is greater than 10, so option C is incorrect.

Process 3: The chromium content is 86.7% and can only contain carbon, hydrogen, and oxygen elements. It is easy to deduce that the compound is Cr_3C_2 through the mass fraction; hydrogen is a reducing atmosphere, and the oxidation product of carbon should be CO . Therefore, the equation is balanced:

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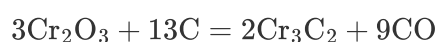
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The compound is Cr_3C_2

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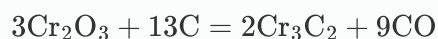
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Hydrogen is a reducing atmosphere, and the oxidation product of carbon should be CO



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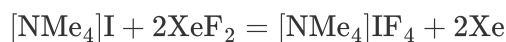
The sum of the coefficients is 27, so option D is incorrect.

Process 4: The reaction occurs at a molar ratio of 1 : 2. Since XeF_2 is a strong oxidizing/fluorinating agent, only -1 valent iodine can be oxidized; it can be known from the molar ratio that IF_4^- is produced, and XeF_2 is reduced to Xe. Balancing the equation:

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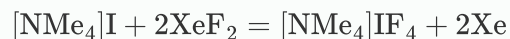
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XeF_2 is a strong oxidizing/fluorinating agent



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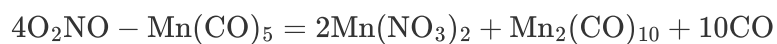
The sum of the coefficients is 6, so option E is incorrect.

Process 5: The common manganese nitrate is $\text{Mn}(\text{NO}_3)_2$. Judging the valence shows that manganese is oxidized; since a decomposition reaction occurs and the nitrate radical still exists, it can only be a disproportionation reaction, so $\text{Mn}(0)$ is produced, and a stable carbonyl compound $\text{Mn}_2(\text{CO})_{10}$ can be conceived; therefore, the equation is balanced:

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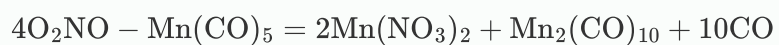
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A decomposition reaction occurs and the nitrate radical still exists, it can only be a disproportionation reaction, so $\text{Mn}(0)$ is produced, and a stable carbonyl compound $\text{Mn}_2(\text{CO})_{10}$ can be conceived



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The sum of the coefficients is 17, which is a prime number, so option F is correct.